# Jack Crane

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## Professional Summary

Software engineer with 6+ years of professional experience and a track record beyond traditional early-career scope, including end-to-end ownership of production enterprise applications and responsibilities typically assigned to senior engineers. Experienced in leading technical direction, building scalable AI and data systems, and partnering with executives and engineering teams to deliver high-impact products.

## Education

**Saint Louis University** St. Louis, Missouri — BA in Computer Science — May 2026

## Experience

### World Wide Technology — Software Engineer & AI Analyst — May 2026 - Present

- Own and develop an enterprise platform for monitoring AI usage, adoption, and spend across providers, models, teams, and users.
- Build production services and data pipelines using Python, Vue.js, MongoDB, PostgreSQL, Snowflake, RabbitMQ, Kafka, Kubernetes, OpenShift, AWS, and GCP.
- Partner with engineering, business, and executive stakeholders to translate requirements into scalable technical solutions.
- Support enterprise deployment and integration of Claude, OpenAI, Copilot, Gemini, Glean, and internally developed AI tools.

### SLU Center for Additive Manufacturing — Software Application Engineer / Research Adjacent — Aug 2022 - Jul 2026

- Lead technical strategy, architecture, and production delivery for SLUCAM's software portfolio, from image and data-processing tools to enterprise project-management systems.
- Architected CoreDesk, a JavaScript/TypeScript project-management tool; advanced it from a proof of concept to production and led five developers through continued design and deployment.
- Led Voxel Print research, reverse-engineering Stratasys full-color 3D printers and publishing a workflow that converts images into print-ready slices.
- Designed a high-volume Go/C data-processing pipeline that converted more than 1.2 billion proprietary 3D drone-scan points into full-color, 3D-printable files.
- Engineered Python-based FuseTools/FuseLapse by reverse-engineering Formlabs TCP and WebSocket protocols and building image-processing workflows for SLS-print timelapses.

### Open Source with SLU — Technical Lead / Senior Software Developer — Jul 2025 - Jun 2026

- Led continued development of CoreDesk, a custom project-management tool, growing it from a narrow-use tool to a configurable platform used by more than 1,000 SLU students, faculty, and staff.
- Recruited, trained, and managed a five-developer team across two semesters, directing delivery, code quality, and the product roadmap.
- Served as the first undergraduate invited to be a Technical Lead in the SLU Open Source program.
- Built a maintainable production stack with Node.js, React, TypeScript, Prisma, PostgreSQL, RabbitMQ, Docker, Jest, Cypress, Cloudflare Workers, DigitalOcean, and AWS.

### Differential Dev Shop — Software Engineer — Oct 2022 - July 2024

- Built a fintech solution that earned Apollos 3 new clients, resulting in $100k+ revenue
- Rapidly grew from IC to system architect and engineering, responsible for CI/CD, deployment, and reliability on a platform with over 1 million MAU
- Built across Python, JavaScript, TypeScript, React Native, and Swift; managed AWS, on-premises, and hybrid infrastructure with Terraform.

### Myers Industries — Freelance developer — Jul 2021 - Nov 2021

- Rapid-prototyped a mobile ordering app, backend, and database that enabled sales representatives to replace consumable hardware quickly after stock outages.
- Generated $3.2 million in 2025 Xpress-order revenue, increasing average line-item count by 400% and average order value by 160%.
- Achieved 60% voluntary adoption among sales representatives while materially reducing the lag between stock outages and replacement orders.

## Passion Projects

### Ohio River Paddlefest — Volunteer Coordinator — 2017 - Present

- Helped deliver a 2026 Guinness World Record for the “Largest Paddle Craft Party,” bringing more than 1,400 people onto the Ohio River.
- Recruited, trained, managed, and supported more than 200 volunteers across the four-day event.

### FIRST Robotics — 2012 - 2022

- Founded and captained high school a team that competed in the state championship 3 of 4 years
- Participated in the world championship.
- Earned the highest individual leadership award in the state
- Mentored/coached an inner city high school team and FTA/Emceed tournaments

## Publications

### Visualizing Machine Learning with a Fluid Simulation [ *https://go.jackcrane.rocks/vmlwfs* ] (Pages 9-11) — March 2025

This paper shows how handwritten numbers can be recognized by simulating how water would flow over their shapes and using those flow patterns to tell digits apart. It uses this physical analogy to make machine learning ideas more visual and intuitive, rather than relying on complex AI models.

### Translating Images into Voxel Print-ready slices [ *https://go.jackcrane.rocks/voxel-print* ] — Jan 2026

This paper describes a workflow for converting standard 2D images into voxel-level, material-specific slice files for Stratasys PolyJet printers, enabling prints that go beyond traditional mesh-based geometry. It combines ICC-based color correction with stochastic halftoning to map image color, transparency, and luminance directly to per-voxel material assignments suitable for Voxel Print systems.